Rule-Based Monetary Policy in China: A Semi-Structural Counterfactual Analysis

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December 19, 2025

Abstract

This proposal outlines a research project aimed at quantifying the effects of rule-based monetary policy in China using a semi-structural approach. Specifically, it investigates how the trajectory of the Chinese economy (Industrial Output, CPI, and PPI) would have differed had the People's Bank of China (PBoC) adhered to a strict target instead of its historical discretionary framework. The study utilizes High-Frequency Identification (HFI) and a modified Romer & Romer approach to recover structural shocks and simulates counterfactuals for the late-Covid period.

1 Background and Motivation

China's economy is currently navigating a significant deflationary cycle. Since the second quarter of 2023, Consumer Price Index (CPI) growth has stagnated near zero, while the Producer Price Index (PPI) has remained in negative territory since late 2022. This price instability coincides with severe liquidity stress in the real estate sector, exemplified by bankruptcy and dangers of default among major developers such as Evergrande and Country Garden and elevated local government debt risks.

Despite the deflationary pressures and looming risks in the financial sector, the People's Bank of China (PBoC) has maintained a cautious, discretionary policy stance. For example, The Loan Prime Rate (LPR), which is the new policy rate set for loans, was lowered only moderately from 3.65% in 2022 to approximately 3.0% recently, a pace arguably insufficient to arrest the deflationary spiral. Meanwhile, although the targets for output growth remained achivable, that for CPI, which was set to 3% for past few years, was consistently missed.

The recent lowering of the 2025 inflation target to roughly 2%, alongside the Central Economic Work Conference's explicit new goal for monetary policy to "promote economic growth and rebound the price level by a reasonable amount" (*wu jia he li hui sheng*), signals a potential regime shift.

2 Research Question

This context creates a critical question to investigate: Was the "discretionary" failure to meet targets a result of structural constraints, or simply a policy choice? Specifically, we aim to analyze the counterfactual the trajectory of the Chinese economy in the late-Covid period (2022-2025), including indexes like GDP, CPI, PPI, and the USD/CNY exchange rate if the PBoC had adhered to a strict, rule-based monetary policy (for instance an explicit 2% Inflation Target or a Nominal GDP Target) instead of its historical discretionary approach

3 Methodology

3.1 The Semi-Structural Approach

We employ the semi-structural methodology developed by McKay and Wolf (2023). This approach bridges the gap between reduced-form empirics and structural theory. Instead of building a fully specified DSGE model, which is prone to misspecification regarding China's complex micro-frictions, we rely on two sufficient statistics to simulate the economy under alternative policy rules:

1. A baseline projection model caputuring the fundamentals and also the shocks (although not identified) of the economy.

2. The causal effects of monetary policy shocks on the macroeconomic variables, representing in Impulse Response Functions (IRFs).

3.2 Key Assumptions

- **Linearity:** We assume the economy's response to shocks is linear (scale and state invariant). While Chen et al. (2025) model regime-switching in the policy rule itself, standard empirical literature (e.g., Das and Song (2022)) supports using linear IRFs for aggregate variables (Industrial Output, CPI) as a robust first-order approximation.

- **Information Symmetry (and its Violation):** A key assumption in the method developed by McKay and Wolf (2023) is that policy shocks are "pure" and do not convey private information about the economic outlook. In China, this assumption is likely violated for two reasons. First, the general public—including the private sector and research institutions—has limited access to government data compared to the authorities, creating significant information asymmetry. Second, the institutional coordination between monetary and fiscal policy means monetary actions

often signal broader fiscal stances or official growth outlooks (Das and Song, 2022). As a result, monetary policy decisions may come with a strong "information effect," causing the transmission to behave differently than expected. For the purpose of this proposal, we will proceed under the full information assumption, but we intend to estimate more robust shock series with alternative identification strategies (such as the advanced HFI methods in He et al. (2024)) in future work to alleviate this limitation as even the shocks themsevels maybe identified but the effects are difficult to disentangled.

- **Single Policy Instrument:** Although McKay and Wolf (2023) did not explicitly state this, but the method relies on that the two casual effects IRFs are derived for the same policy instrument, which is the federal funds effective rate. However, in China, the PBoC targeted both the qantity of money (via Reserve Requirement Ratio, RRR) and the price of money (via 7-day repo rate, R007) simultaneously. Only in recent years has the PBoC shifted more towards the interest rate-based policy framework. For now, we will start with measuring the effects of R007-based shocks (which Chen et al. (2025) also did for the 2000-2018 period). However, if the results are not robust, we may shift the focus to the RRR-based shocks (as in Chen et al. (2018)) and evaluate the policy counterfactuals during the stock market crisis (2015-2016).

3.3 Baseline Projection

The original work by McKay and Wolf (2023) used a 10-variable system for estimating the baseline projection using reduced-form VAR in the U.S. context, including variables like unemployment, output gap, investment, consumption, working hours, Utilization-adjusted TFP, labor productivity, labor share and inflation. However, in the China context, many of these variables are not available or reliable and also there are some other indexes specific to China. The following is the list adjusted to fit China's economy, dual-track price system:

1. **GDP:** The Nomial GDP data by Chen et al. (2024) to avoid the over-smoothiness in the official data (China Cyclical Activity Tracker (CCAT) Fernald et al. (2020) for robustness checks).
2. **Consumer Prices:** CPI Inflation (YoY).
3. **Producer Prices:** PPI Inflation (YoY).
4. **Policy Instrument:** 7-Day Repo Rate (R007).
5. **External Constraint:** CNY/USD Exchange Rate (Capturing "Impossible Trinity" constraints).

6. **Financial Friction:** Corporate Bond Spread (e.g., AA-AAA spread or LGFV spread).

3.4 Casual Effects of Monetary Policy series 1: High-Frequency Identification (HFI)

To obtain the causal IRFs of monetary policy shocks, we need to identify the shocks. The first approach we employ is similar to the HFI series constructed by Das and Song (2022):

- **Instrument:** 7-day Repo Rate (R007) changes on PBoC policy announcement days instead of the 1-year Interest Rate Swaps (IRS) in order to unify the policy instrument with the baseline projection model and the Romer & Romer shock which will be discussed later.

- **Event Window:** [-1, +1] days around major policy announcements (including policy interest rate changes, RRR changes, open market operations, foreign exchange rate regime reform, release of quarterly China Monetary Policy Reports (PBoC), annual Reports on the Work of the Government (State Council), annual speeches by the Chairman on the Central Economic Work Conference). In addition, we am considering include policy leaks prior to the announcement by the foreign media (e.g., Reuters, Bloomberg and WSJ). If data allows, we will also test a more narrower window of half day before and after the announcement.

An Alternative approach for HFI is proposed by Bu et al. (2021) and later adopted by He et al. (2024) to the China context. This method improves identification by first estimating the sensitivity of the entire yield curve (swaps and bonds) to policy events, and then using these sensitivities to filter daily rate changes. This process extracts a single latent shock series that isolates the common policy signal from idiosyncratic market noise more effectively than single-instrument proxies.

3.5 Casual Effects of Monetary Policy series 2: Romer & Romer Shocks

Another way to recover the historical monetary policy shocks ($\epsilon_{R,t}$), we estimate a "China-version" of the Romer and Romer shock (Romer and Romer (2004)), modifying by reaction function from Chen et al. (2025), which 1. dropped the use of internal forecasts (which is not publicly available in China) and 2. have not yet included the narrative component extracting from the official documenets. The estimated Rule is as follows:

$$R_t = \alpha + \gamma_R R_{t-1} + \gamma_\pi(\pi_t - \pi^*) + \gamma_{x,t}(y_{x,t} - y^*) + \epsilon_{R,t} \tag{1}$$

R_t is the 7-day repo rate (R007), π_t is the quarterly YoY CPI, and the $y_{x,t}$ is the quarterly YoY real GDP grwoth rate while y^* is the corresponding annual grwoth rate target set in the annual Report on the Work on the Government. The residual $\epsilon_{R,t}$ represents the monetary policy shocks. We also caputure the assymetric reponse to the output gap similar to Chen et al. (2025) as follows:

$$\gamma_{x,t} = \begin{cases} \gamma_{x,a} & \text{if } g_{x,t} \geq g^*_{x,t} \\ \gamma_{x,b} & \text{if } g_{x,t} < g^*_{x,t} \end{cases}, \tag{2}$$

Unlike Chen et al. (2025), who use quarterly data and a fixed target on CPI (annual 3.5% and 0.85% quarterly) , we use quarterly YoY data and include the targets set each set(π^*) to match the changes in policy stance. Also, their work allows for the state-dependent heteroskedasticity in the policy but this paper does not incorporate this for now, which might complicate the linearity assumption.

3.6 The Rest of the Procedure

Once the monetary policy shocks are identified, the analysis proceeds in two stages. First, we will estimate the dynamic causal effects of these shocks on the key macroeconomic variables to obtain the Impulse Response Functions (IRFs) using VAR. These IRFs serve as the "sufficient statistics" capturing the economy's transmission mechanism. Second, utilizing the algorithm from McKay and Wolf (2023), we will construct the counterfactual projections. This involves solving for the specific sequence of interest rate changes that would have been required to satisfy the alternative strict targeting rule (e.g., maintaining a 2% inflation target) at every point in time, holding the historical sequence of non-monetary shocks fixed. By comparing the counterfactual projections to the actual outcomes, especially CPI, GDP, and exchange rate, we can see how the PBoC were trading off between those indexes.

4 Some Preliminary Results

We have successfully replicate the shocks identification designed by Chen et al. (2025) for the 2001-2023 period. Noted that in 2020 the government did not set an explicit GDP growth target among the uncertainties brought by COVID-19 pandemic, so we set the target to be 6%, which fit the trend but may affect the estimation. The estimated monetary policy rule is plotted in Figure 1.

From the figure, we notice that the shocks (the difference between the actual and fitted lines) shows some patterns. Specifically, the fitted line is often a right-shift of the actual line. This phenomennon can be attributed to 1. the autocorrelation among the variables which can be expected from the lag term and 2.the fact that PBoC might react

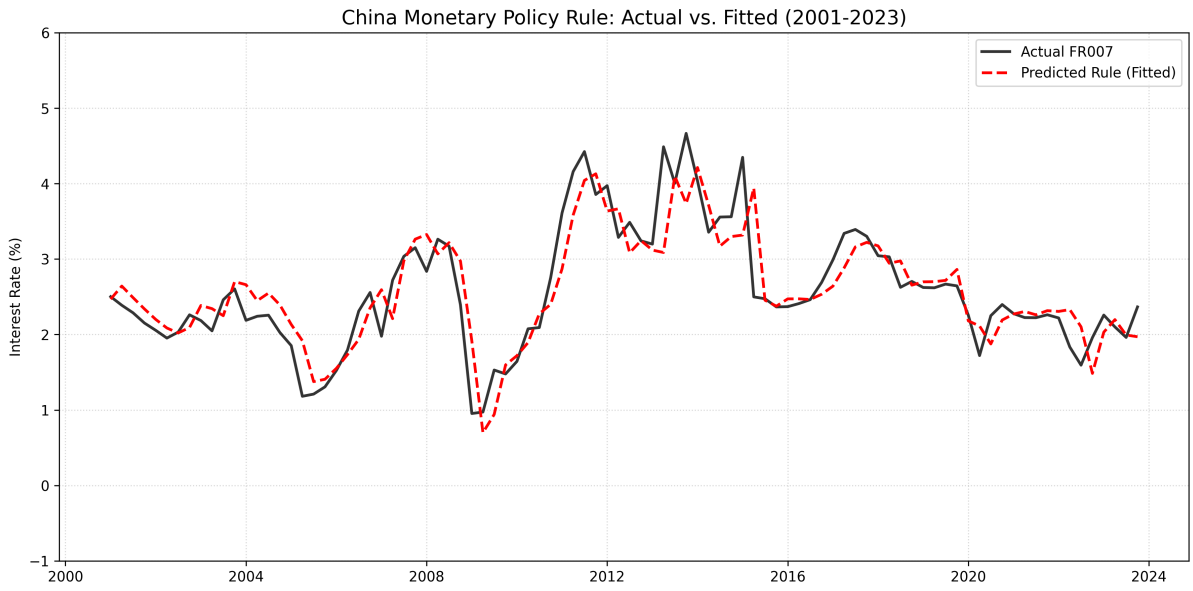

Figure 1: Estimated Monetary Policy Rules for China (2001-2023)

to the market faster or even preemptively. To investigate this further, we re-estimate the rule without the lagged dependent variable. Although the "right-shift" disappears, the result is not be econometrically sound as the errors terms are not dealt with properly. The results are plotted in Figure 2.

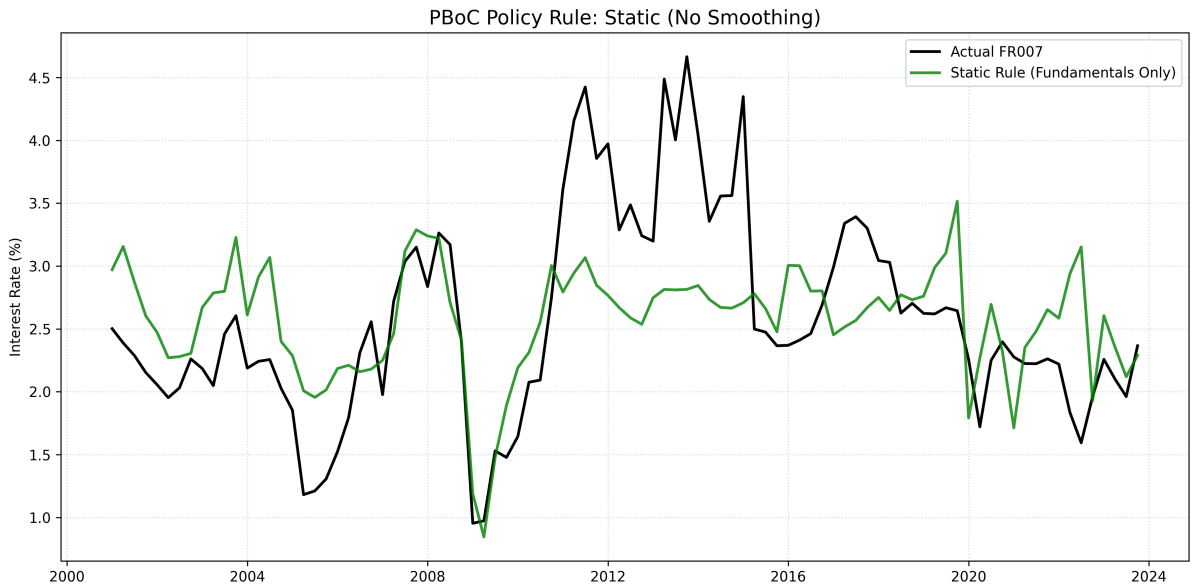

Figure 2: Estimated Monetary Policy Rules for China (No Lagged Dependent Variable)

References

Bu, Chunya, John Rogers, and Wenbin Wu. (2021). A Unified Measure of Fed Monetary Policy Shocks. *Journal of Monetary Economics*, 118 (March), 331-349.

Caravello, T. E., McKay, A., & Wolf, C. K. (2025). Evaluating Monetary Policy Counterfactuals: (When) Do We Need Structural Models? *Working Paper*.

Chen, K., Ren, J., & Zha, T. (2018). The Nexus of Monetary Policy and Shadow Banking in China. *American Economic Review*, 108(12), 3891–3936.

Chen, K., Patrick Higgins, and Tao Zha (2024). Constructing Quarterly Chinese Time Series Usable for Macroeconomic Analysis. *Journal of International Money and Finance*, 143 (2024) 103052.

Chen, K., Xiao, Y., & Zha, T. (2025). A Trade-off Between Monetary Policy Transmission and Systemic Risk in China. *NBER Working Paper 34056*.

Das, S., & Song, W. (2022). Monetary Policy Transmission and Policy Coordination in China. *IMF Working Paper WP/22/74*.

He, J., Jia, D., Li, K., & Wu, W. (2024). A High-Frequency Measure of Chinese Monetary Policy Shocks. *PHBS Working Paper*.

McKay, A., & Wolf, C. K. (2023). What Can Time-Series Regressions Tell Us About Policy Counterfactuals? *Econometrica*, 91(5), 1695–1725.

Fernald J., Eric Hsu, Mark M. Spiegel (2021). Is China fudging its GDP figures? Evidence from trading partner data. *Journal of International Money and Finance*, 110 (2021), 102262.

Romer, Christina D., and David H. Romer. (2004). A New Measure of Monetary Shocks: Derivation and Implications. *American Economic Review*, 94(4), 1055-1084.